Ashmit Dutta

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Education

University of Illinois Urbana-Champaign — Urbana, IL
Bachelor of Science, Computer Engineering — *Aug 2023 – May 2027*

- Accepted paper at ICML 2025 workshop on finetuning Chain of Thought in LLMs

Experience

Nablon AI — Jun 2026 – Present
ML Engineering Intern — *Remote — San Francisco, CA*

- Building automated eval pipelines to surface failure modes between RL runs before promotion.

Assured and Trustworthy AI Research Lab – University of Illinois — Dec 2025 – Present
ML Research Intern (Prof. Huan Zhang) — *Urbana, IL*

- Shipping Python and C++ features to **auto_LiRPA** verification library, VNN-COMP 2021–2025 winner.
- Building evaluation pipelines and benchmark datasets for the lab's VNN-COMP 2026 global submission.

John Deere — Jan 2025 – Jan 2026
Software Engineer Intern — *Champaign, IL*

- Shipped a React + Spring Boot data-exporting platform **to production** for **500+ internal users**.
- Built core REST APIs and CI/CD pipelines over PostgreSQL and DB2 handling **millions of records**.
- Designed a multithreaded job scheduler for search queries in the DB **cutting search time down 3x**.
- Added AWS CloudWatch observability cutting issue triage time, and secured sensitive export data in S3.

University of Chicago Data Science Institute — DSI Summer Lab — Jun 2024 – Aug 2024
ML Research Intern (Fermilab collaboration) — *Chicago, IL*

- Selected for DSI's competitive summer research program; built a reusable multi-modal Graph Neural Network library in PyTorch Geometric for Fermilab's Exa.TrkX neutrino-reconstruction project.
- Engineered an end-to-end training pipeline over **20,000+ detector events** (H5Py) on a Slurm HPC cluster; presented results at the DSI summer research symposium.

Projects

Stanford TreeHacks: Multi-Modal Video Search Engine | *Python (Modal, BrightData, FastAPI)* — Feb 2026

- **Winner (Modal Inference Track + Bright Data AI) at Stanford TreeHacks 2026** (out of 380 teams).
- Built a distributed full-stack video search and dataset-generation app on cloud GPU inference.
- Designed a multi-modal embedding pipeline fusing vision and audio for semantic retrieval, indexing 24 hr+ videos.

GPU Accelerated Deep Learning Inference (LeNet-5) | *C++, CUDA* — Feb 2026 – Apr 2026

- Wrote shape-specialized implicit-GEMM CUDA kernels with fused im2col and constant-memory weights.
- Accelerated Conv2 with WMMA Tensor Cores for **6.6× speedup**; 12.5 ms inference at batch 10,000.
- Drove design via Nsight Compute roofline/stall analysis; **placed 5/130 in ECE 408 competition**.

Autonomous Drone Racing | *ROS, Python, C++, PyTorch* — Oct 2025 – Dec 2025

- Built a ROS 2 stack for a Crazyflie 2.1 drone using a min-snap planner and geometric controller.
- Tuned controller gains via Bayesian optimizer; **outperformed 3 published baselines** on real hardware.
- Trained a YOLOv8 gate-segmentation model on NeRF-rendered data (**98% accuracy**).
- Analyzed flight Rosbags to debug tracking error; designed a voltage-battery feedback system to prevent thrust sag.

Leadership, Publications & Awards

Co-Founder & Lead Organizer, Online Physics Olympiad (OPhO): largest student-run international physics competition — built a **22k+ Discord community**, 3k+ Facebook following, and **3k peak annual competitors** since 2020; raised **$10k+ in sponsorships** from Citadel, Jane Street, and Wolfram.

Awards & Leadership: 2nd Place, Bradley Mottier Innovation Challenge, UIUC ISE ($2,000) — AR glasses for ASL translation. Executive Board Member (Treasurer), IEEE Chapter at UIUC.

Technical Skills

Languages: Python, C/C++, Java, JavaScript, SQL, SystemVerilog, RISC-V Assembly
ML & AI: PyTorch/Geometric, CUDA, Modal, LLM APIs, embeddings, multi-modal retrieval
Backend & Infra: REST APIs, FastAPI, Spring Boot, PostgreSQL, DB2, MongoDB, Docker, Git, CI/CD, AWS, Linux
Systems & Hardware: RTL design, FPGAs, STM32/PCB design, embedded firmware, GDB, QEMU